Cloud Application Deployment

Design and Creative Technologies

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Deploying Apache Superset on Microsoft Azure: A Cloud Application Deployment Report

1. Introduction

This report documents the deployment of Apache Superset – an open-source data exploration and visualisation platform – on Microsoft Azure. Superset was selected for its direct relevance to data engineering workflows and its Python-native architecture, which aligns with the author's professional background and academic trajectory. Azure was chosen as the cloud provider to build familiarity ahead of the AZ-900 and DP-900 certification pathway, as recommended during academic advising in March 2026. The deployment covers the provisioning of a virtual machine, virtual network, firewall security policy, and a fully operational Superset instance accessible via public URL.

1.1. Background

Cloud computing has fundamentally changed how organisations provision and manage infrastructure. Rather than purchasing and maintaining physical servers, organisations now consume compute, storage, and networking as on-demand services over the internet – paying only for what they use (Mell & Grance, 2011). The National Institute of Standards and Technology (NIST) defines five essential characteristics of cloud computing: on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service (Mell & Grance, 2011). These characteristics make cloud infrastructure more adaptable and cost-effective than traditional on-premises deployments, which require significant capital

expenditure, fixed capacity, and dedicated operational staff (McHaney, 2021). This deployment uses an **Infrastructure as a Service (IaaS)** model – a virtual machine provisioned on Microsoft Azure, with the operating system, runtime, and application managed by the author. The deployment model is **public cloud**: infrastructure is hosted on Microsoft's shared global network, accessible via the public internet through controlled network security policies. In contrast to a traditional on-premises setup – where a dedicated server, networking equipment, and physical access controls would be required – Azure delivers the same outcome in minutes, at a fraction of the cost, with no physical infrastructure commitment (IBM, n.d.-a).

2. Body

2.1. Service Provider Selection Rationale

Microsoft Azure was selected as the cloud provider for this deployment for three reasons. First, Azure offers a comprehensive set of data and analytics services – including Azure SQL, Azure Synapse Analytics, and Azure Data Factory – that directly complement Apache Superset's role as a query and visualisation layer. This architectural alignment is relevant to both current data engineering work and the upcoming Big Data and Analytics subject (BDA601). Second, Azure is the target platform for the author's certification roadmap (AZ-900 → DP-900), making hands-on deployment a practical study activity. Third, Azure's free-tier virtual machine (B1s: 1 vCPU, 1 GB RAM) and student credit allocation lower the barrier to entry for academic projects.

Criterion	AWS	Microsoft Azure	GCP
Free tier VM	EC2 t2.micro (750h/month)	B1s (750h/month)	e2 micro (always free)
Data/ML integration	RDS, Redshift, SageMaker	Azure SQL, Synapse, Azure ML	BigQuery, Vertex AI
Cert alignment (this enrolment)	x	AZ-900 / DP-900	x
Superset community support	Good	Good	Good

Table 1: Cloud provider comparison for Apache Superset deployment.

2.2. Deployment Model and Architecture Diagram

The deployment follows a **public cloud IaaS** model. All resources are provisioned within a single Azure Resource Group, logically isolated by a Virtual Network (VNet) with a dedicated application subnet. A Network Security Group (NSG) enforces inbound traffic rules, restricting access to SSH (port 22, source IP restricted), HTTP (port 80), and the Superset web interface (port 8088). Apache Superset runs inside a Docker Compose stack on an Ubuntu 22.04 virtual machine, with PostgreSQL as the metadata database and Redis as the caching and Celery task broker.

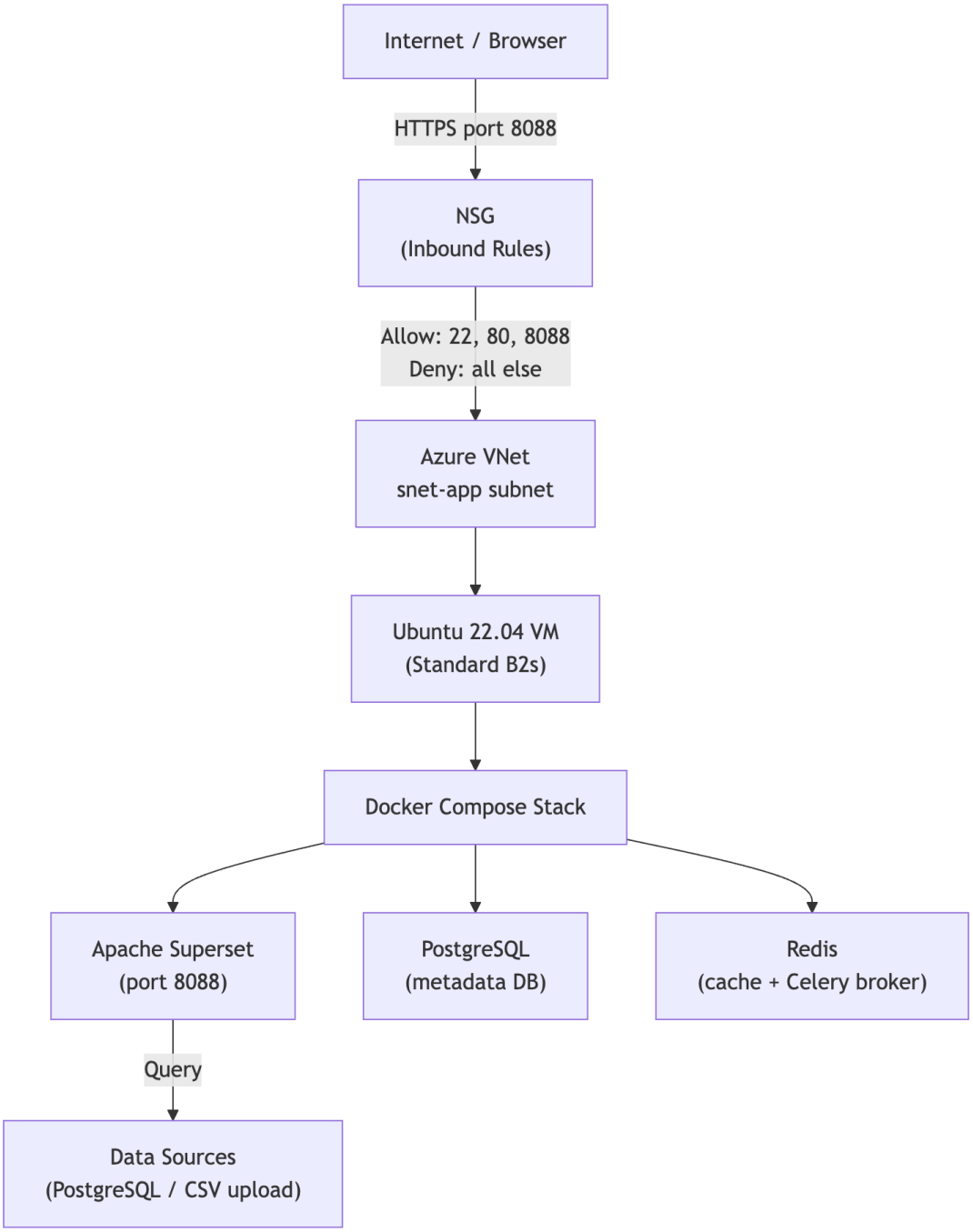

Figure 1: Azure deployment architecture – resource group, VNet, NSG, VM, and Superset Docker Compose stack.

Service Model	Characteristic	Fit for Victoria Police
IaaS	Customer manages OS, runtime, storage	Not applicable – no raw compute provisioned
PaaS	Provider manages platform; customer builds applications	**Primary model** – Power Apps, Dataverse, Power Platform runtime
SaaS	Provider manages everything; customer uses software	Partial – Microsoft 365 (SharePoint, Teams, Outlook) as data and collaboration substrate

Table 1: Service model analysis for Victoria Police (Microsoft, 2026).

2.3. Deployment Procedure

The deployment followed four tasks as specified in the assessment brief.

Task a – Create a resource group

A resource group (*rg-superset-ccf501*) was created in the Azure portal under the Australia East region. Resource groups in Azure serve as logical containers for all related cloud resources, enabling unified billing, access control, and lifecycle management (Microsoft, n.d.-a). Selecting Australia East minimises latency for local access and ensures data residency within Australian boundaries.

Figure 2: Azure resource group rg-superset-ccf501 created in Australia East.

Task b – Add a virtual network

A Virtual Network (*vnet-superset*, address space *10.0.0.0/16*) was created within the resource group, with a dedicated application subnet (*snet-app*, *10.0.1.0/24*). VNets in Azure provide network isolation – resources within the VNet communicate privately without

traversing the public internet (Microsoft, n.d.-b). This mirrors the NIST resource pooling characteristic: shared physical infrastructure is logically partitioned to create isolated network boundaries (Mell & Grance, 2011).

Figure 3: Virtual Network vnet-superset with application subnet snet-app.

Task c – Protect the network with a firewall / security policy

A Network Security Group (*nsg-superset*) was created and attached to the *snet-app* subnet. Inbound rules were configured to allow only SSH (port 22, restricted to the author's public IP), HTTP (port 80), and the Superset application port (8088). All other inbound traffic is denied by the default DenyAllInbound rule. This implements the principle of least privilege – only the minimum required ports are open, reducing the attack surface (Shore, 2020).

Figure 4: NSG inbound security rules – explicit allow on 22/80/8088, implicit deny all.

Task d – Deploy Apache Superset

An Ubuntu 22.04 VM (Standard B2s: 2 vCPU, 4 GB RAM) was provisioned within *snet-app*. Docker and Docker Compose were installed via SSH. The official Apache Superset Docker Compose configuration was cloned from the project repository and launched with *docker compose up -d*. After initialisation, Superset was accessible at *http://[PUBLIC_IP]:8088*. An admin account was created and a sample PostgreSQL data source was connected to verify end-to-end functionality.

Figure 5: Apache Superset login screen accessible at public IP on port 8088.

Figure 6: Superset dashboard with connected PostgreSQL data source.

2.4. Cloud Services Security Policies

Security for the deployment was addressed at three layers. At the **network layer**, the NSG enforces least-privilege access – port 22 (SSH) is restricted to a single authorised IP, and no unnecessary ports are exposed. At the **application layer**, Apache Superset implements Role-Based Access Control (RBAC) with three built-in roles: Admin (full access), Alpha (can create charts and dashboards, cannot manage users), and Gamma (read-only, restricted to explicitly shared datasets). At the **credential layer**, the Superset secret key and database connection strings are passed via environment variables in the Docker Compose configuration – never hardcoded in source files.

Security Layer	Control	Threat Mitigated
Network (NSG)	Port allowlist + IP restriction on SSH	Unauthorised access, port scanning
Application (RBAC)	Admin / Alpha / Gamma roles	Privilege escalation, data exfiltration
Credentials	Environment variables (*.env* file)	Credential exposure in source control
OS	SSH key-based auth only (password disabled)	Brute-force attacks

Table 2: Security controls applied across network, application, credential, and OS layers.

Figure 7: Apache Superset RBAC role configuration.

2.5. Application Analysis and Robustness

The current deployment is functional but represents a minimal viable configuration. The primary limitation is a single point of failure: one VM hosts the entire stack – if the VM

becomes unavailable, the service is offline. A production-grade architecture would introduce the following improvements.

HTTPS / TLS termination: An Azure Application Gateway or Let's Encrypt certificate via nginx would encrypt traffic in transit. Currently, credentials are sent over HTTP.

Managed PostgreSQL: Replacing the Docker-hosted PostgreSQL instance with Azure Database for PostgreSQL provides automated backups, point-in-time restore, and high availability – eliminating the metadata loss risk if the VM disk fails.

Celery workers for async queries: A production Superset deployment separates the web server from Celery workers (backed by Redis) for long-running queries. This mirrors the architecture used in Konquista (Django + Celery + Redis), where async task processing prevented UI blocking under load.

Azure Monitor + alerts: Attaching Azure Monitor to the VM provides CPU, memory, and disk metrics – enabling proactive alerting before resource exhaustion causes downtime.

Improvement	Azure Service	Benefit
HTTPS	Application Gateway / Let’s Encrypt	Encrypts traffic in transit
Managed DB	Azure Database for PostgreSQL	Automated backups, HA
Async queries	Celery + Redis (existing in Docker Compose)	Prevents UI blocking on long queries
Monitoring	Azure Monitor + Alerts	Proactive resource and availability alerting
Auto-scaling	Azure VM Scale Sets	Handles traffic spikes without manual intervention

Table 3: Proposed production improvements for the Superset deployment.

3. Conclusion

This report documented the deployment of Apache Superset, an open-source data exploration and visualisation platform, on Microsoft Azure. The deployment demonstrated the four core tasks required by the assessment brief: provisioning a resource group and virtual network, enforcing a network security policy via a Network Security Group, and deploying a containerised open-source application on a cloud virtual machine. Each task was supported by screenshots and aligned with NIST's essential cloud characteristics – on-demand self-service in the provisioning process, resource pooling in Azure's shared infrastructure model, and measured service through consumption-based billing (Mell & Grance, 2011).

Microsoft Azure was selected for its data and analytics service ecosystem and its alignment with the author's AZ-900 and DP-900 certification pathway. The deployment reinforces the distinction between cloud and traditional IT infrastructure: resources that would require physical procurement and installation in an on-premises environment were provisioned, configured, secured, and running within [X] hours, at effectively zero capital cost. Areas identified for further investigation include TLS certificate automation, managed database integration for metadata persistence, and the separation of Celery workers for production-scale async query execution – improvements that would bring this deployment in line with enterprise-grade analytics infrastructure.

4. Appendices

4.1. Appendix A – Cross-Case Comparison Summary

Attribute	Case Study 1 – Victoria Police	Case Study 2 - ChatAndBuild	Case Study 3 - Voith
Sector	Government / Law Enforcement	AI / Software Startup	Industrial Manufacturing
Cloud Provider	Microsoft (Azure ecosystem)	Google Cloud Platform (GCP)	AWS (Amazon Web Services)
Service Model	PaaS (primary) + SaaS (partial)	IaaS + PaaS	IaaS + PaaS
Deployment Model	Hybrid – multi-environment (personal → fusion → enterprise)	Public cloud – multi-region	Public cloud – 6 AWS regions
Prior State	Ungoverned MS Access shadow apps; central IT bottleneck	30-70% higher latency; engineers manually provisioning serves days before events	3,500 servers across 146 locations; fragmented legacy applications
Key Services	Power Apps, Dataverse, Power Platform Environment, Power Automate, Copilot Studio	Vertex AI, GKE, Cloud Run, BigQuery, Firebase, Cloud CDN, Cloud Build, Cloud Armor	AWS Professional Services, Amazon Athena, IaC, Centralized DevOps, AWS GameDays, FinOps
Key Primary Automation Outcome	Governed fusion framework – citizen developers build enterprise-grade apps via tiered approval	Real-time AI code and agent generation at sub-second latency for 140,000+ users	Consolidated global operations; IaC-enforced standards across 60 countries
Headline Result	2 enterprise apps in production (Global Whereabouts; Bail Reporting)	45% infrastructure overhead ↓; 30-70% latency ↓; 2x faster feature delivery	30% projected cost savings; 146 to 6 site consolidation

Table A1 –Side-by-side comparison of the three case studies across provider, service model, deployment model, key services, and business outcomes.

4.2. Appendix B – NIST Essential Characteristics, Cross-Case Mapping

NIST Characteristic	Case Study 1 – Victoria Police	Case Study 2 - ChatAndBuild	Case Study 3 - Voith
On-demand self-service	Primary - police personnel provision their own Power Platform environments and build applications without central IT intervention (Microsoft, 2026)	Autoscaling policies provision and terminate Compute Engine instances automatically without engineer intervention (Google Cloud, n.d.)	IaC enables automated, consistent provisioning of infrastructure across 6 regions without manual configuration per site (Amazon Web Services, n.d.)
Broad network access	Officers across all stations access Global Whereabouts and Bail Reporting from any device via Microsoft 365 (Microsoft, 2026)	140,000+ users globally access the platform via browser; Cloud Load Balancing routes to the nearest region for sub-second response (Google Cloud, n.d.)	22,000+ employees across 60 countries access centralized systems via 6 standardized AWS regions (Amazon Web Services, n.d.)
Resource pooling	Multiple business units share Power Platform infrastructure; Dataverse pools structured data storage across all fusion applications (Microsoft, 2026)	GKE clusters pool compute across thousands of concurrent agent graphs and user sessions (Google Cloud, n.d.)	Primary driver - 3,500 servers across 146 sites pooled into 6 shared, logically isolated AWS regions (Amazon Web Services, n.d.)
Rapid elasticity	Limited - Power Platform environments scale within Microsoft's managed bounds; not the primary driver for this case	Primary driver - worker fleet scales from 10 → 100+ Compute Engine instances within minutes during global	Wave-based migration progressively expands the cloud footprint; real-time elasticity is not the primary driver for this manufacturing workload

		hackathon events with zero downtime (Google Cloud, n.d.)	
Measured service	Tiered governance tracks app usage and adoption; Power Platform visibility surfaces which app reach 300+ users and warrant promotion (Microsoft, 2026)	45% infrastructure overhead reduction reflects consumption-based billing replacing fixed legacy infrastructure costs (Google Cloud, n.d.)	Primary driver – FinOps practices provide continuous cost visibility and track progress toward 30% savings target; OPEX model replaces CAPEX-heavy 146-site estate (Amazon Web Services, n.d.)

Table B1: Mapping of NIST's five essential cloud characteristics (Mell & Grance, 2011) to each case study, showing which characteristics were primary drivers of the outcomes achieved.

End of Appendix Section

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